

MATH40005 Spring Midterm

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Part A

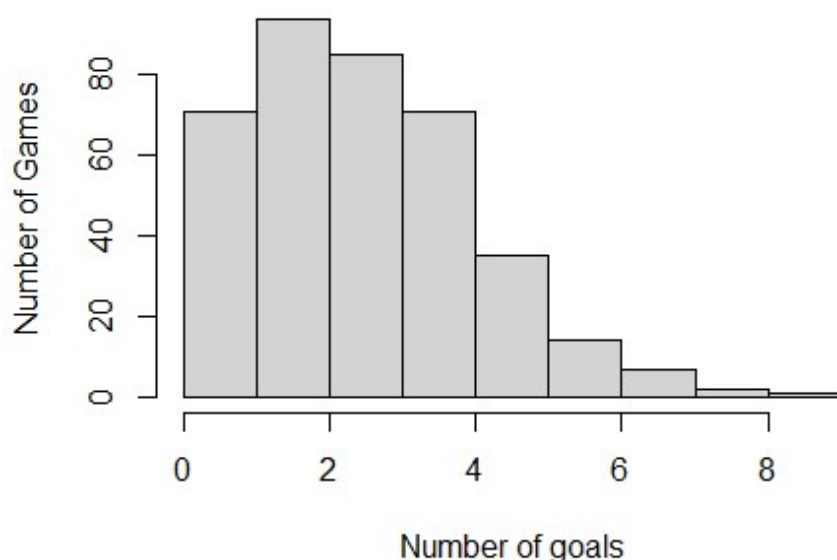
```
df <- read.table('PL2425.txt', sep=',', header=T)
# Checking data imported correctly by checking first three lines
head(df,3)
```

##	Round	Match	Day	Date	Time	HomeTeam	Score	AwayTeam
## 1	1	1	Fri	16/08/2024	20:00	Manchester United	1-0	Fulham
## 2	1	2	Sat	17/08/2024	12:30	Ipswich Town	0-2	Liverpool
## 3	1	3	Sat	17/08/2024	15:00	Newcastle	1-0	Southampton
##	HomeGoals	AwayGoals	Attendance	Venue				
## 1	1	0	73297	Old Trafford				
## 2	0	2	30014	Portman Road Stadium				
## 3	1	0	52196	St James Park				

Part B

```
numGoals <- df$HomeGoals + df$AwayGoals
hist(numGoals, xlab='Number of goals', ylab='Number of Games',
main='Histogram showing number of goals per game')
```

Histogram showing number of goals per game



Part C

```
#Isolate Fulham Home Games
Fulham_home <- df$HomeTeam == 'Fulham'
# Find attendances at these games
attFulham <- df[Fulham_home, 'Attendance']
# Find median attendance
attFulham_median <- median(attFulham)
cat('Median attendance at Fulham home matches:',attFulham_median)

## Median attendance at Fulham home matches: 26954
```

Part D

```
# T test with 18 degrees of freedom. Finding critical t values
n <-length(attFulham)
L <- qt(0.025, n)
U <- qt(0.975, n)
#Sample variance of fulham attendance
Ful_samp_var <- var(attFulham)
Ful_samp_mean <- mean(attFulham)
# Calculate lower bound and upper bound
LBF <- L*sqrt(Ful_samp_var)/sqrt(n) +Ful_samp_mean
UBF <- U*sqrt(Ful_samp_var)/sqrt(n) +Ful_samp_mean
cat('95% confidence interval:', LBF, UBF)

## 95% confidence interval: 26449.78 27201.69

#Checking D Using t.test
# Use ttest function
t.test(attFulham)

##
## One Sample t-test
##
## data: attFulham
## t = 149.34, df = 18, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 26448.36 27203.11
## sample estimates:
## mean of x
## 26825.74
```

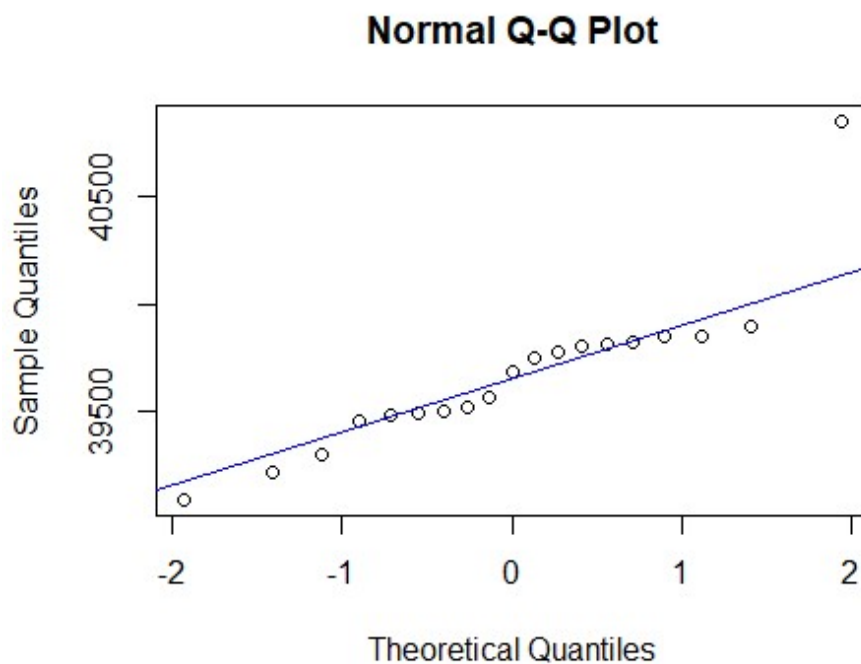
Part E

H0: $\mu_1 = \mu_2$

H1: μ_1 not equal to μ_2

Part F

```
#Isolate Chelsea Home games  
Chelsea_home <- df$HomeTeam == 'Chelsea'  
# Find attendances at these games  
attChelsea <- df[Chelsea_home, 'Attendance']  
# Use qqnorm  
qqnorm(attChelsea)  
qqline(attChelsea, col='blue')
```



The data seems to follow a straight line suggesting Chelsea's attendance is normally distributed. However, the lowest 3 points and upper 3 points vary from line by a fair amount. This plot is therefore inconclusive; this may be because there are only 19 data points. If Chelsea were to play more games at home, perhaps a conclusion could be reached.

Part G

Assumes that random variables generating samples have the same variance i.e. $\sigma_1^2 = \sigma_2^2$

```
# Investigate Arsenal's attendance variance. Then investigate Chelsea's  
attendance variance  
#Isolate Arsenal Home games  
Arsenal_home <- df$HomeTeam == 'Arsenal'  
# Find attendances at these games
```

```

attArsenal <- df[Arsenal_home, 'Attendance']
#Calculate sample variance of Arsenal
Ars_samp_var <- var(attArsenal)
#Calculate sample variance of Chelsea
Che_samp_var <- var(attChelsea)
cat("The sample variance of Chelsea's home attendance is:", Che_samp_var,
"The sample variance of Arsenal's home attendance is:", Ars_samp_var )

## The sample variance of Chelsea's home attendance is: 135639 The sample
variance of Arsenal's home attendance is: 7174.708

#Creating a 95% confidence interval for each team's attendance variance using
chisquared test with 18 degrees of freedom.
lower <- qchisq(0.025,18)
upper <- qchisq(0.975,18)
n <- length(attChelsea)-1
#Chelsea confidence interval
LBC <- (n-1)*Che_samp_var/upper
UBC <- (n-1)*Che_samp_var/lower

#Arsenal confidence interval
LBA <- (n-1)*Ars_samp_var/upper
UBA <- (n-1)*Ars_samp_var/lower
cat('95% confidence interval for Chelsea variance:', LBC, UBC)

## 95% confidence interval for Chelsea variance: 73140.77 280152.4

cat('95% confidence interval for Arsenal variance:', LBA, UBA)

## 95% confidence interval for Arsenal variance: 3868.825 14818.83

```

Confidence intervals for Chelsea and Arsenal attendance variance don't overlap so the assumption they are the same is rejected